Republic of the Philippines

BALIWAG POLYTECHNIC COLLEGE

(Dalubhasaang Politekniko ng Lungsod ng Baliwag — BTECH)

Baliwag, Bulacan

INSTITUTE OF INFORMATION TECHNOLOGY AND INNOVATION (IITI)

Bachelor of Science in Information Technology

OUTCOMES-BASED COURSE SYLLABUS

First Semester, Academic Year 2025–2026

SAMPLE DOCUMENT — prepared for AILAS testing; not an official BTECH syllabus.

I. INSTITUTIONAL PHILOSOPHY, VISION, MISSION AND GOALS

Philosophy

Guided by belief in God and humanitarian principles, Baliwag Polytechnic College supports the State policy of providing affordable, quality education to all its citizens.

Vision

To be at the forefront of the community's desire to motivate its people to pursue knowledge; to be an instrument of social transformation; and to preserve national history and culture while harnessing technology and keeping pace with international demands.

Mission

To provide relevant and updated quality educational development to the underprivileged youths of Baliwag and its neighboring towns.

Institutional Goals

1. Ensure the alignment of the plans, programs and activities of the academic departments and offices with the vision, mission and objectives of the College.
2. Infuse the culture of research, community service, peace, information technology, and skills training and development into all curricula as prescribed by CHED, DepEd and TESDA.

II. INSTITUTE VISION, MISSION AND OBJECTIVES (IITI)

Vision

To be the top choice for quality IT education for the youth, producing efficient problem solvers, researchers, system innovators, and competent ICT professionals.

Mission

To provide relevant, quality IT education and services to students, graduates, and faculty.

Objectives

- Recruit qualified and competent faculty who serve as role models.
- Offer quality IT degree programs that develop management and technical skills.
- Produce competent IT graduates responsive to the needs of industry.
- Take an active part in community service.
- Conduct significant research in information technology.

IITI Graduate Attributes (INFOTECH)

Innovative problem solvers · Networked professionals · Faculty role models · Outstanding management skills · Technical proficiency · Efficiency · Community engagement · High-quality research

III. COURSE INFORMATION

Course Code	ITC 212
Course Title	Data Structures and Algorithms
Credit Units	3 units (2 hours lecture, 3 hours laboratory per week)
Year Level / Semester	2nd Year, 1st Semester

Pre-requisite	ITC 121 — Computer Programming 2
Curriculum	BSIT Curriculum 2022 (aligned with CHED CMO No. 25, s. 2015)
Schedule / Room	MW 1:00–2:00 PM (Lec) · F 1:00–4:00 PM (Lab) · IT Lab 2
Faculty	[Faculty Name], [Degree] — [email]@btech.edu.ph
Consultation Hours	Tuesday & Thursday, 3:00–5:00 PM, IITI Faculty Room

IV. COURSE DESCRIPTION

This course introduces the fundamental data structures and algorithms used to organize, store and process data efficiently. Students analyze algorithm complexity using Big-O notation and implement arrays, linked lists, stacks, queues, recursion, trees, heaps, hash tables and graphs, together with classic searching and sorting algorithms. Emphasis is placed on choosing the appropriate data structure for a given computing problem and on writing correct, efficient and well-documented code through hands-on laboratory work and a team mini-project.

V. BSIT PROGRAM OUTCOMES (CHED CMO NO. 25, S. 2015)

Upon completion of the BSIT program, graduates are able to:

Code	Program Outcome
PO-a	Apply knowledge of computing, science and mathematics appropriate to the discipline.
PO-b	Analyze a problem and identify and define the computing requirements appropriate to its solution.
PO-c	Design, implement and evaluate computer-based systems, processes, components or programs to meet desired needs.
PO-d	Function effectively on teams to accomplish a common goal.
PO-e	Understand professional, ethical, legal, security and social issues and responsibilities.
PO-f	Communicate effectively with a range of audiences.
PO-i	Use current techniques, skills and tools necessary for computing practice.
PO-l	Recognize the need for, and engage in, independent and life-long learning.

VI. COURSE LEARNING OUTCOMES (CLO)

At the end of the course, students should be able to:

Code	Course Learning Outcome	Mapped POs
CLO1	Analyze the time and space complexity of algorithms using Big-O notation.	a, b
CLO2	Implement linear data structures (arrays, linked lists, stacks and queues) to solve computing problems.	a, c, i
CLO3	Apply recursion and divide-and-conquer techniques in algorithm design.	a, b
CLO4	Implement and compare searching and sorting algorithms based on efficiency.	b, c, i
CLO5	Construct non-linear data structures (trees, heaps, hash tables and graphs) and their traversal algorithms.	a, c, i
CLO6	Design, document and present a team software solution that selects appropriate data structures for a real-world problem.	c, d, f, l

VII. COURSE COMPETENCIES AND TOPICS

Each competency below is assessed in the grading period indicated and is the unit used for mastery tracking (75% mastery threshold).

Code	Competency / Topic	CLO	Period	Learning Outcome
DSA-01	Algorithm Analysis and Big-O Notation	CLO1	Prelim	Determine the growth rate of an algorithm; compare O(1), O(log n), O(n), O(n log n) and O(n^2).
DSA-02	Arrays and Linked Lists	CLO2	Prelim	Implement singly, doubly and circular linked lists; perform insertion, deletion and traversal.

DSA-03	Stacks and Queues	CLO2	Prelim	Implement stacks and queues using arrays and linked lists; evaluate postfix expressions.
DSA-04	Recursion and Divide-and-Conquer	CLO3	Midterm	Write recursive solutions; trace the call stack; identify base and recursive cases.
DSA-05	Searching Algorithms	CLO4	Midterm	Implement linear and binary search; justify when binary search is applicable.
DSA-06	Sorting Algorithms	CLO4	Midterm	Implement bubble, selection, insertion, merge and quick sort; compare their complexity and stability.
DSA-07	Trees and Binary Search Trees	CLO5	Semi-Final	Build a BST; perform in-order, pre-order and post-order traversal; insert and delete nodes.
DSA-08	Heaps and Priority Queues	CLO5	Semi-Final	Implement a binary heap; apply heap sort and priority-queue operations.
DSA-09	Hash Tables	CLO5	Semi-Final	Design hash functions; resolve collisions through chaining and open addressing.
DSA-10	Graphs and Graph Traversal	CLO5	Final	Represent graphs using adjacency lists and matrices; implement BFS and DFS.
DSA-11	Shortest Path and Minimum Spanning Trees	CLO5	Final	Apply Dijkstra's, Prim's and Kruskal's algorithms to weighted graphs.
DSA-12	Data Structure Selection and Applied Design	CLO6	Final	Choose and justify data structures for a real-world system; document and present the solution.

VIII. LEARNING PLAN

Wk	Topic	Competency	Teaching–Learning Activity	Assessment
1	Orientation; course VMGO; review of Programming 2	—	Lecture, diagnostic test	Diagnostic quiz (ungraded)
2	Algorithm analysis; Big-O notation	DSA-01	Lecture, code tracing	Quiz 1
3	Arrays; singly linked lists	DSA-02	Lab: linked list operations	Lab Exercise 1
4	Doubly and circular linked lists	DSA-02	Pair programming	Lab Exercise 2
5	Stacks and queues; postfix evaluation	DSA-03	Lab: expression evaluator	Quiz 2, Lab Exercise 3
6	PRELIMINARY EXAMINATION	DSA-01–03	—	Prelim Exam
7	Recursion; tracing the call stack	DSA-04	Lecture, whiteboard tracing	Quiz 3
8	Linear and binary search	DSA-05	Lab: search benchmarking	Lab Exercise 4
9	Elementary sorts (bubble, selection, insertion)	DSA-06	Lab: sorting visualizer	Lab Exercise 5
10	Merge sort and quick sort	DSA-04, DSA-06	Lecture, complexity comparison	Quiz 4
11	MIDTERM EXAMINATION	DSA-04–06	—	Midterm Exam
12	Trees; binary search trees; traversals	DSA-07	Lab: BST implementation	Lab Exercise 6
13	Heaps and priority queues; heap sort	DSA-08	Lecture, lab	Quiz 5
14	Hash tables; collision resolution	DSA-09	Lab: phonebook with hashing	Lab Exercise 7
15	SEMI-FINAL EXAMINATION	DSA-07–09	—	Semi-Final Exam
16	Graphs; BFS and DFS	DSA-10	Lab: barangay road map traversal	Lab Exercise 8
17	Dijkstra's, Prim's and Kruskal's algorithms	DSA-11	Lecture, problem solving	Quiz 6
18	Mini-project presentation; FINAL EXAMINATION	DSA-10–12	Team presentation	Final Exam, Project

IX. ASSESSMENT AND GRADING SYSTEM

Each grading period grade is computed from the following components:

Component	Weight	Type
Quizzes	20%	Formative

Laboratory Exercises	30%	Formative
Major Examination (Prelim / Midterm / Semi-Final / Final)	40%	Summative
Recitation and Class Participation	10%	Formative

Final Grade = (Prelim 20%) + (Midterm 20%) + (Semi-Final 20%) + (Final 40%). The team mini-project (CLO6) is graded as the Final-period laboratory output.

Percentage	Grade	Remarks
97–100	1.00	Excellent
94–96	1.25	Excellent
91–93	1.50	Very Good
88–90	1.75	Very Good
85–87	2.00	Good
82–84	2.25	Good
79–81	2.50	Satisfactory
76–78	2.75	Satisfactory
75	3.00	Passed
Below 75	5.00	Failed
—	INC	Incomplete requirements

X. COURSE POLICIES

- **Attendance:** Students who incur absences exceeding 20% of the total class hours may be dropped from the course, per the Student Handbook.
- **Academic honesty:** Plagiarism, code copying and cheating in any assessment result in a grade of zero for that requirement and referral to the Office of Student Affairs.
- **Late submission:** Laboratory outputs submitted late incur a 10% deduction per day, up to three days.
- **Missed examinations:** A special examination may be given only for valid, documented reasons filed within one week.
- **Early intervention:** Students with three or more competencies below 75% mastery, or a grade decline of more than 15% between grading periods, are scheduled for consultation and provided remedial resources.
- **Data privacy:** Student records are handled in accordance with the Data Privacy Act of 2012 (RA 10173).

XI. REFERENCES

- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to Algorithms (4th ed.). MIT Press.
- Goodrich, M. T., Tamassia, R., & Goldwasser, M. H. (2014). Data Structures and Algorithms in Java (6th ed.). Wiley.
- Sedgewick, R., & Wayne, K. (2011). Algorithms (4th ed.). Addison-Wesley.
- Bhargava, A. Y. (2016). Grokking Algorithms. Manning Publications.
- VisuAlgo — visualising data structures and algorithms. https://visualgo.net

XII. PREPARED, REVIEWED AND APPROVED

Prepared by:	*Reviewed by:*	*Approved by:*
_____	_____	_____
[Faculty Name]	**[Program Director Name]**	**[Dean Name]**
Instructor, IITI	Program Director, BSIT	Dean, Institute of Information Technology and Innovation
Date: _____	Date: _____	Date: _____